

VERIFICATION OF KIRCHHOFF'S LAWS

AIM:

To verify kirchhoff's voltage law and kirchhoff's current law both theoretically and practically for a given DC circuit.

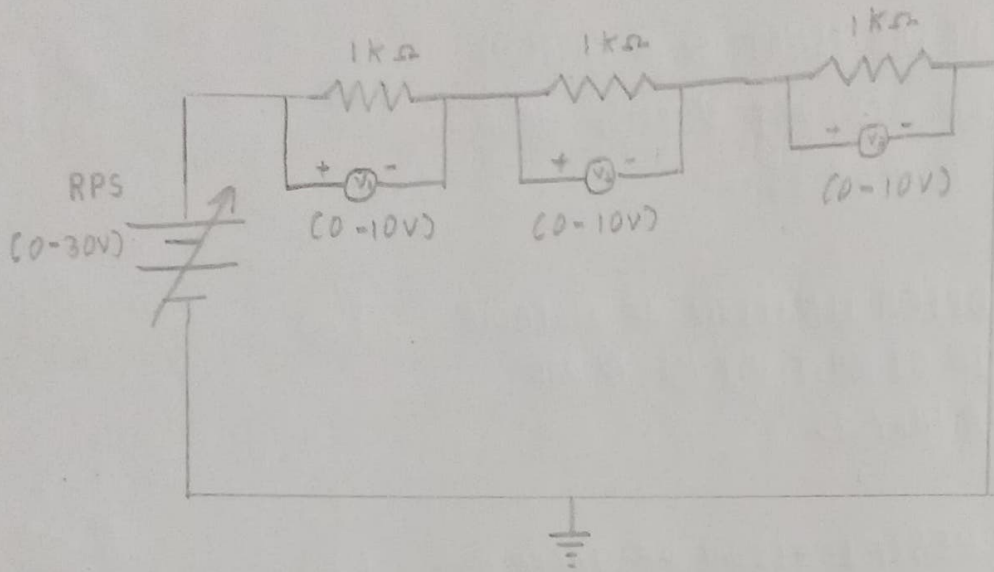
APPARATUS REQUIRED:

SL. NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power supply (RPS)	(0-30V)	1
2.	Voltmeter	(0-30V) MC	3
3.	Ammeter	(0-10mA) MC	3
4.	Resistors	1 k Ω	3
5.	Bread board	--	1

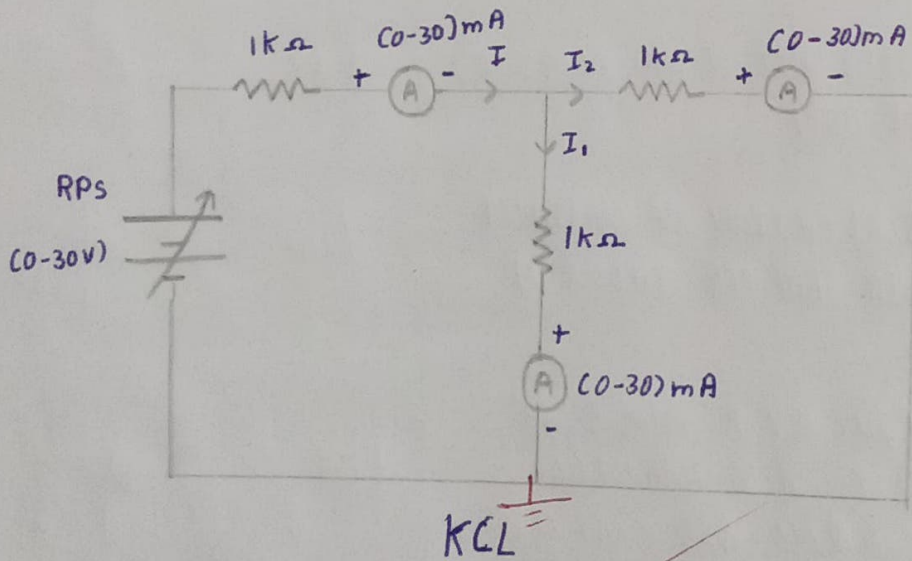
PROCEDURE:

- 1) Connect the circuit as per the diagram.
- 2) Switch ON the RPS and set the input voltage
- 3) Note the readings of Voltmeters and ammeters.
- 4) Vary the input voltage and record all readings
- 5) Switch off the RPS.
- 6) Verify kirchhoff's laws using the recorded values.

CIRCUIT DIAGRAM :



KVL



8

KCL

TABULAR COLUMN:

V (volts)	V ₁ (volts)	V ₂ (volts)	V ₃ (volts)	V = V ₁ + V ₂ + V ₃ (volts)	I (amps)	I ₁ (amps)	I ₂ (amps)	I ₁ + I ₂ (amps)
2	0.7	0.7	0.7	2.1	1.16	0.58	0.58	1.16
6	2.03	2.03	2.03	6.09	3.4	1.7	1.7	3.4
9	3.05	3.05	3.05	9.15	4.8	2.4	2.4	4.8

all verified

CALCULATIONS:

• Practical Verification:

V/I should be constant (ohm's law)

V should be equal to (V₁ + V₂ + V₃)

I₁ should be equal to (I₂ + I₃)

• Theoretical Verification:

Kirchoff's Voltage Law:

i) for V = 2 volts:

$$V_1 = 0.7, V_2 = 0.7, V_3 = 0.7$$

$$V = V_1 + V_2 + V_3 = 0.7 + 0.7 + 0.7 = 2.1V \approx 2V$$

ii) for V = 6 volts:

$$V_1 = 2.03, V_2 = 2.03, V_3 = 2.03$$

$$V = V_1 + V_2 + V_3 = 2.03 + 2.03 + 2.03 = 6.09 \approx 6V$$

iii) for V = 9 volts:

$$V_1 = 3.05, V_2 = 3.05, V_3 = 3.05$$

$$V = V_1 + V_2 + V_3 = 3.05 + 3.05 + 3.05 = 9.15 \approx 9V$$

- Kirchhoff's current law:

$$\text{for } V=2, I=1.16A, I_1=0.58A, I_2=0.58A$$

$$I = I_1 + I_2 = 0.58 + 0.58 = 1.16A$$

$$\text{for } V=6, I=3.4A, I_1=1.7A, I_2=1.7A$$

$$I = I_1 + I_2 = 1.7 + 1.7 = 3.4$$

$$\text{for } V=9, I=4.8A, I_1=2.4A, I_2=2.4A$$

$$I = I_1 + I_2 = 2.4 + 2.4 = 4.8A.$$

MARKS SPLIT - UP:

SL. NO	Description	Marks Allotted	Marks Obtained
1	Experimental Setup / Connections	20 marks	20
2.	observation / Readings	20 marks	20
3.	Calculations / Results	20 marks	20
4.	Viva	20 marks	20
5.	Record	20 marks	20
		Total	100

100
10/2/20

RESULT:

Thus ~~Kirchoff's~~ Voltage law and Kirchoff's current law are verified partially and theoretically.